

Projecting MLB Bat Speed

with Measurement Error Models, Missing Data Imputation, and Gaussian Processes

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Abstract We propose a framework to forecast Major League Baseball (MLB) bat speed using Statcast data. Because Statcast bat speed data only spans from mid-2023 to the present (summer 2026 upon writing), we extended the dataset back to 2016 by imputing missing values from the conditional distribution of bat speed given 90th percentile exit velocity (EV90). To account for sample-size volatility in monthly exit velocity, we implemented a measurement error model to estimate each player’s latent EV90. This latent metric denoised observed performance, yielding a stable anchor for historical imputation and long-term projection.

Introduction

A hitter’s contact quality and bat speed determine exit velocity, a primary key to offensive success in baseball. In this paper, we forecast bat speed independently of contact quality, plate discipline, and batted-ball variance. To capture how bat speed evolves over age, we implemented an additive Gaussian Process model with player-level random effects.

MLB’s Statcast dataset provided pitch-level observations, which we aggregated to the player-month level. However, because Statcast has only tracked bat speed since mid-2023, standard forecasting methods suffer from the narrow performance window. To address this missing data, we developed a two-step framework: first, estimate each player’s true underlying exit velocity through a measurement error model, and second, impute missing bat speed values via a multivariate normal distribution.

In the first step, the measurement error model, we estimated a player’s latent 90th percentile exit velocity (EV90). Monthly observed EV90 is often unreliable for players with few balls in play (BIPs) and prior literature indicates exit velocity requires roughly 40 BIPs to stabilize. For example, utility player Luke Williams averaged only 12.1 BIPs per month between July 2021 and April 2023, causing his observed monthly EV90 to fluctuate between 76 and 103 mph. By modeling observed EV90 as a function of latent EV90 plus measurement error, we established a reliable baseline for each player.

In the second step, missing data imputation, we imputed missing bat speed observations using the relationship between bat speed and latent EV90. Physics literature establishes exit velocity as a function of bat speed and collision efficiency [1]. Because Statcast does not record collision efficiency, we estimated the joint multivariate normal distribution of bat speed and latent EV90, then sampled missing bat speed values from the conditional distribution $p(BS \mid \theta_{EV90}^{\text{true}})$. Methodologically, this conditional sampling operates as a Bayesian regression of bat speed on latent EV90, with missing values informed by the underlying correlation between the two metrics.

Exploratory Data Analysis

Before specifying our model, we outline our data transformation and cleaning pipeline. We downloaded pitch-level data using the pybaseball package and scraped player biographical information from MLB’s API. We removed non-regular-season games and set bat speed values below 50 mph to null, as these likely reflected tracking errors on non-competitive swings such as bunt attempts. For exit velocity, we restricted the dataset to balls in play, retaining home runs and excluding bunt attempts.

We aggregated the cleaned observations to the player-month level, calculating average bat speed and 90th percentile exit velocity (EV90). We combined March with April and September with October, yielding six monthly observations per season: Mar/Apr, May, Jun, Jul, Aug, and Sep/Oct. Additionally, we removed player-months with an EV90 below 65 mph (fewer than 0.5% of observations) to prevent non-competitive swings from distorting imputed bat speed values.

The below image displays these monthly metrics alongside their respective sample sizes. As shown, EV90 exhibits larger variance at lower ball-in-play counts, whereas bat speed stabilizes much faster.

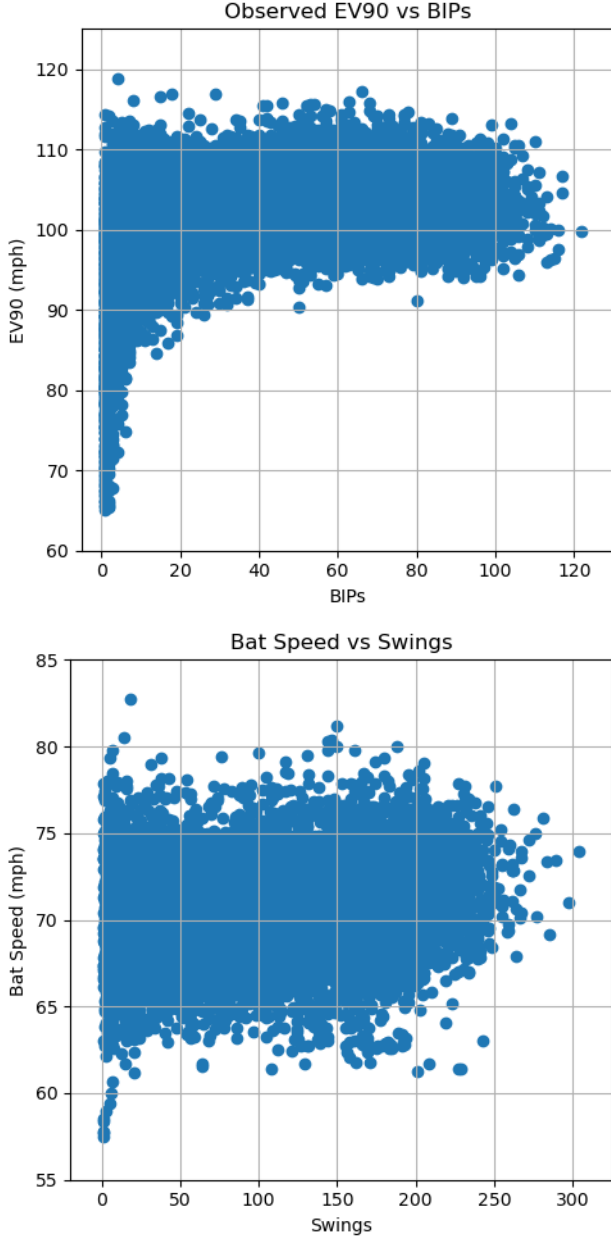


Figure 1: Sample size stability for monthly EV90 (top) and average bat speed (bottom)

We extracted player position labels from the MLB API. Because these labels represent a player’s primary or last known position, the labels do not capture positional shifts given a time frame. For example, Mookie Betts was listed as a shortstop despite also playing second base and right field. For simplicity, we ignored any positional shifts. Table 1 summarizes the unique player counts and monthly observations across each positional group. Besides the standard position labels, we also grouped some positions into three broader categories: Up the Middle (UTM), Corner Outfielders (COF), and Corner Infielders (CINF). The UTM category includes shortstops, second basemen, and center fielders, reflecting common defensive transitions across middle-of-the-field positions. COF includes left fielders, right fielders, and generic outfielders, while CINF comprises first and third basemen to capture the common shift of third basemen moving to first base later in their careers.

While useful, this classification scheme is inherently imperfect. For instance, right fielders occasionally transition to first base, and catchers sometimes move to 1B. Although player profiles are better defined by underlying hitting metrics than static defensive labels, we opted for this simpler positional grouping instead of minor league data, where raw offensive metrics require competition level adjustments (e.g., a .400 slugging percentage in Double-A does not translate directly to a .400 slugging percentage in MLB).

Finally, we computed each player’s monthly age, defined as their age on the 15th of each month. Across all observations, the sample exhibited a mean age of 28.2 ± 3.7 years.

Position	Position Group	Unique Players	Observations
DH	DH	59	1,815
RF	COF	180	2,848
OF	COF	30	314
LF	COF	217	3,349
3B	CINF	183	3,214
1B	CINF	177	3,338
SS	UTM	188	3,116
CF	UTM	198	2,886
2B	UTM	211	3,488
C	C	270	4,636

Table 1: Distribution of unique players and monthly observations by position

Measurement Error Model

Monthly observed EV90 exhibited noise driven by sample-size volatility. Some months a player put everything in play, and in other months, they were ice cold. In months where recorded EV90 fell below 95 mph, the median sample size was only three BIPs. Because exit velocity metrics require roughly 40 BIPs to stabilize, imputing bat speed directly from raw EV90 observations would skew downstream forecasting. To isolate underlying performance, we implemented a measurement error model to estimate each player’s true latent EV90, denoted as $\theta_{t,p}^{\text{true}}$.

Standard measurement error frameworks specify $x^{\text{obs}} \sim N(x^{\text{true}}, \epsilon)$, where x^{obs} represents the observed measurement and x^{true} represents underlying latent skill. We extended this framework in two key ways. First, we scaled observation variance inversely by monthly BIPs to reflect greater statistical certainty in larger samples. Second, we modeled latent EV90 as a Gaussian random walk over time, allowing player skill to drift continuously across monthly periods rather than treating each month independently. Formally, for player p at time t , the model is specified as:

$$\begin{aligned}
\theta_{t,p}^{\text{obs}} &\sim N(\theta_{t,p}^{\text{true}}, \frac{\sigma_{\epsilon}^2}{\text{bipst}_{t,p}}) \\
\theta_{t,p}^{\text{true}} | \theta_{t-1,p}^{\text{true}} &\sim N(\theta_{t-1,p}^{\text{true}}, \sigma_{\text{month}}^2) \\
\theta_{0,p}^{\text{true}} &\sim N(\mu, \sigma_p^2) \\
\mu &\sim N(\hat{\mu}, \hat{\sigma}_{\mu}^2) \\
\sigma_{\epsilon} &\sim \text{Exp}(0.1) \quad \sigma_p \sim \text{Exp}(1) \quad \sigma_{\text{month}} \sim \text{Exp}(3)
\end{aligned}$$

The first line defines our measurement likelihood, where ob-

served EV90 is normally distributed around latent EV90 with variance inversely proportional to monthly BIPs. The second line models latent skill evolution as a random walk governed by monthly variance σ_{month}^2 . The third line specifies the initial state prior, drawing each player’s baseline skill from a population distribution with mean μ and between-player variance σ_p^2 . The fourth line places an empirical prior on μ , and the fifth places exponential priors on all standard deviation parameters. We selected these prior scales after conducting prior predictive checks. We fit the model via Markov Chain Monte Carlo using the No-U-Turn Sampler (NUTS) in PyMC. All parameters achieved convergence with $\hat{R} \leq 1.01$.

Posterior estimates yielded a between-player standard deviation (σ_p) of 2.6 mph and a monthly innovation standard deviation (σ_{month}) of 0.3 mph, indicating that latent skill moves gradually across a season. The posterior mean for observation noise (σ_ϵ) was 12.5, representing single-play variance. Because observation noise scales down by $\sqrt{bips_{t,p}}$, single-observation variance shrinks rapidly as sample sizes grow, explaining the high volatility in low-BIP months. Posterior predictive checks confirmed alignment between simulated distributions and observed EV90 metrics.

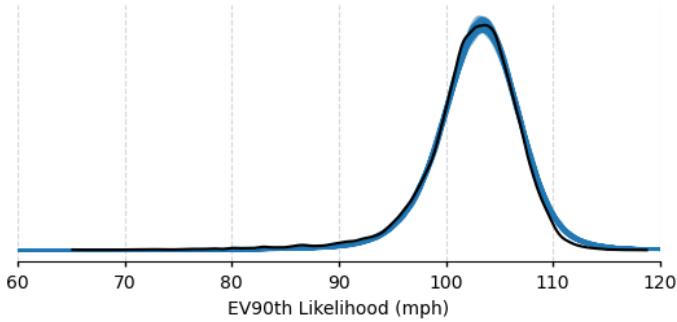


Figure 2: Posterior Predictive of Observed EV90

To quantify the divergence between observed and latent EV90, we computed the difference between the two metrics and averaged it across ball-in-play counts. Figure 3 displays these differences, where the shaded band represents one standard deviation of the difference.

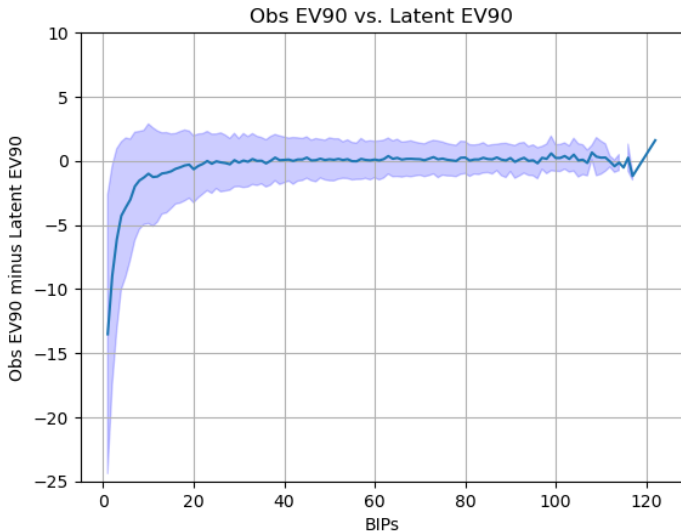


Figure 3: Difference between Obs EV90 and Latent EV90

To examine individual career trajectories, we plotted observed EV90 (red) alongside estimated latent EV90 (blue) for select players. We included red error bars representing the observational standard error ($\sigma/\sqrt{bips}$) and shaded the latent EV90 within a one-standard-deviation interval. Consequently, months with sparse data exhibited longer error bars, while the latent trajectories provided a stabilized estimate.

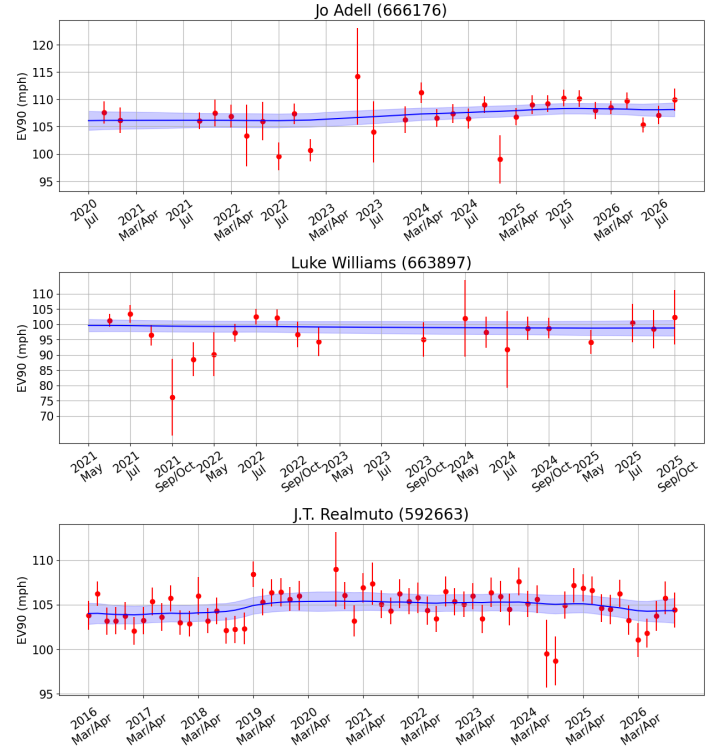


Figure 4: Career Latent EV90

For players like Jo Adell and Luke Williams, who had streaky months during the non-bat-tracking era, the model recognized the underlying noise and smoothed their estimates into a more plausible latent trajectory. By contrast, for regulars like J.T. Realmuto, the observed data was already stable, so the model barely made an adjustment. Finally, in months with zero recorded BIPs, the latent EV90 leaned entirely on the random walk prior and player baseline.

Missing Data Imputation

We first fit a joint multivariate normal (MVN) distribution between latent EV90 and observed bat speed, restricting the training sample to player-months with at least 40 BIPs and 10 swings. Even though latent EV90 is smoother than raw EV90, estimates under 40 BIPs still carried more residual variance than we wanted. For bat speed, 10 swings was sufficient since swing speed stabilizes quickly. We treated the historical missingness as missing completely at random (MCAR) and imputed missing bat speed values by sampling directly from the conditional MVN distribution.

Across our monthly sample, the empirical correlation between latent EV90 and observed bat speed was 0.80, giving an $R^2 = 0.64$. This comes in lower than David Marshall’s swing-level finding that bat speed explains roughly 81% of the variation in exit velocity [2]. Marshall’s model operated at the

pitch level, calibrating hitter-specific collision efficiency and assuming a uniform bat size, both of which directly influence collision physics, while attributing remaining variance to contact quality.

Formally, we model the joint distribution of latent EV90 and bat speed as a multivariate normal:

$$\begin{aligned} \begin{bmatrix} \theta_i^{true} \\ BS_i \end{bmatrix} &\sim N\left(\begin{bmatrix} \mu_\theta \\ \mu_{BS} \end{bmatrix}, \Sigma\right) \\ \Sigma = LL^T &= \begin{bmatrix} \sigma_\theta^2 & \rho\sigma_\theta\sigma_{BS} \\ \rho\sigma_\theta\sigma_{BS} & \sigma_{BS}^2 \end{bmatrix} \\ \mu_\theta &\sim N(\hat{\mu}_\theta, 1) \quad \mu_{BS} \sim N(\hat{\mu}_{BS}, 1) \\ L &\sim \text{LKJCholeskyCov}(\eta = 2, \sigma = \hat{\sigma}^2) \end{aligned}$$

For player-months without bat-tracking data, we imputed missing values directly from the conditional distribution. To propagate uncertainty from both the covariance matrix and our latent EV90 estimates, each posterior draw proceeded in two steps. First, we sampled the latent EV90 for player p in time period t :

$$\theta_{t,p}^{true} \sim \mathcal{N}(\hat{\theta}_{t,p}^{true}, \hat{\sigma}_{\theta,t,p}^2)$$

Then, at the observation level, we drew the imputed bat speed:

$$BS_i^{imp} \sim \mathcal{N}\left(\mu_{BS} + \frac{L_{10}}{L_{00}}(\theta_i^{true} - \mu_\theta), L_{11}^2\right)$$

We fit the model using Markov Chain Monte Carlo via the No-U-Turn Sampler (NUTS) in PyMC, with all parameters achieving convergence at $\hat{R} \leq 1.01$.

Posterior sampling yielded a correlation of 0.87 (95% HDI: [0.86, 0.88]) between latent EV90 and bat speed. Because sampling from the conditional MVN is equivalent to regressing bat speed on latent EV90, the ratio $L_{10}/L_{00} = 0.93$ gives us an intuitive slope that holding all else constant, every 1 mph increase in a player's monthly latent EV90 corresponds to a 0.93 mph bump in expected bat speed. Evaluating the imputation against observed data produced a root mean squared error of 1.36 mph. Visualizing the fitted relationship reveals a slight non-linearity in the lower tail, where for players with a latent EV90 below 100 mph, the linear model overestimates expected bat speed.

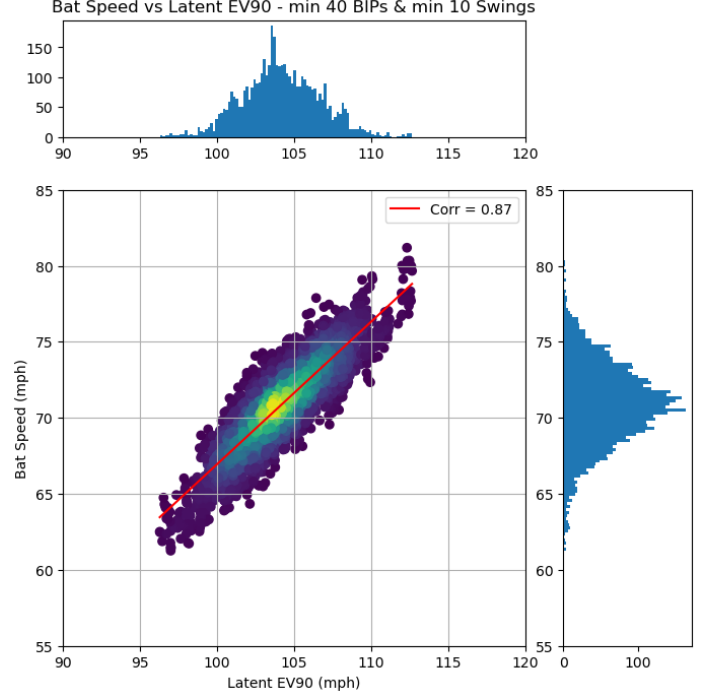


Figure 5: Bat Speed and Latent EV90 Results

Projecting Bat Speed

Finally, right? To construct our complete bat speed dataset, we used observed bat speed whenever available and substituted our conditional imputed values for unobserved months. Consequently, monthly bat speeds from 2016 through mid-2023 rely on imputed estimates, whereas data from mid-2023 to present reflects direct tracking observations.

We modeled bat speed projections using two additive components: a hierarchical player-level random intercept with partial pooling to shrink noisy player baselines toward the population mean, and an aging curve component. We initially specified the aging component as a quadratic polynomial, later extending it to a non-parametric Gaussian process to better capture career trajectory.

Hierarchical Random Effects

We included individual hierarchical random effects as varying intercepts to partially pool player baselines toward the population mean, shrinking small-sample noise. We set an empirical global prior centered at our sample average:

$$\mu_{pop} \sim \mathcal{N}(\hat{\mu}_{BS}, \hat{\sigma}_{BS}^2)$$

where $\hat{\mu}_{BS} = 69.8$ mph and $\hat{\sigma}_{BS}^2 = 2.6$ represent the empirical mean and variance of observed bat speed. Each player's latent intercept was then drawn from the population distribution:

$$\mu_p \sim \mathcal{N}(\mu_{pop}, \sigma_p^2) \quad \sigma_p \sim \text{HalfNormal}(1)$$

where σ_p represents the between-player standard deviation.

After confirming baseline MCMC convergence, we expanded the hierarchical structure to account for positional differences. Defensive positions like center field and shortstop skew toward faster runners with less power, whereas first base and designated hitter attract power hitters with higher natural bat

speeds. To capture these shifts without introducing unidentifiability against the grand mean μ_{pop} , we added positional offsets $\mu_{position}$ subject to a sum-to-zero constraint across all K positions:

$$\begin{aligned}\mu_p &\sim \mathcal{N}(\mu_{pop} + \mu_{position[k]}, \sigma_p^2) \\ \mu_{position[k]} &\sim \mathcal{N}(0, \sigma_{position}^2) \quad \text{subject to} \quad \sum_{k=1}^K \mu_{position[k]} = 0 \\ \sigma_{position} &\sim \text{HalfNormal}(1)\end{aligned}$$

where k indexes the primary position corresponding to player p .

Age Component

To incorporate how bat speed evolves over time, we first modeled the population-level aging trajectory using a quadratic function, $f = a(\text{age})^2 + b(\text{age}) + c$, where age is standardized. To maintain identifiability between the player-level random intercepts and the global aging curve, we fixed $c = 0$.

To capture a non-linear trajectory, we extended this to a Gaussian process: $f \sim \mathcal{GP}(m(\text{age}), k(\text{age}, \text{age}'))$. Rather than assuming a zero mean function, we parameterized $m(\text{age})$ using a quadratic mean function informed by our preliminary model estimates to anchor prior expectations in regions with sparse data. The covariance kernel $k(\text{age}, \text{age}')$ governs similarity across ages. We selected an Exponentiated Quadratic (Radial Basis Function) kernel, where the lengthscale parameter l dictates the distance over which aging trends remain correlated, and the amplitude parameter η scales the magnitude of deviations around $m(\text{age})$. To bypass the $\mathcal{O}(N^3)$ computational bottleneck of full covariance matrix inversion, we implemented a Hilbert Space Gaussian Process (HSGP) approximation using basis functions. The RBF kernel is defined as:

$$k(\text{age}, \text{age}') = \eta^2 \exp\left(-\frac{(\text{age} - \text{age}')^2}{2l^2}\right)$$

Combing it together

Combining these components, we evaluated three model specifications:

1. Quadratic Aging Curve Model: Models bat speed evolution using a quadratic aging curve with player-level random intercepts.
2. HSGP Aging Curve Model: Replaces the quadratic trajectory with a Hilbert Space Gaussian Process aging curve.
3. HSGP & Position Model: Incorporates position-level offsets alongside player-level random effects within the HSGP aging framework.

Model 1: Quadratic Aging Curve Model

We defined the quadratic aging curve model as follows:

$$\begin{aligned}y_{p,i} &\sim N\left(\theta_{p,i}, \frac{\sigma_{m,i}^2}{\text{swings}_{p,i}}\right) \\ \theta_{p,i} &= \mu_p + f_i \\ \mu_p &\sim N(\mu_{pop}, \sigma_p^2) \quad \mu_{pop} \sim N(69.8, 2.6) \quad \sigma_p \sim \text{HalfNormal}(1) \\ f_i &= a \cdot \text{age}_i^2 + b \cdot \text{age}_i \quad a, b \sim N(0, 1) \\ \sigma_{m,i}^2 &= \begin{cases} \sigma_{obs}^2 & \text{if } y_{p,i} \text{ is observed} \\ \sigma_{imp}^2 & \text{if } y_{p,i} \text{ is imputed} \end{cases} \\ \sigma_{obs} &\sim \text{HalfNormal}(1) \quad \sigma_{imp} \sim \text{HalfNormal}(1)\end{aligned}$$

Model 2: HSGP Aging Curve Model

We extended the baseline specification by replacing the quadratic aging curve with a Hilbert Space Gaussian Process (HSGP) aging curve:

$$\begin{aligned}y_{p,i} &\sim N\left(\theta_{p,i}, \frac{\sigma_{m,i}^2}{\text{swings}_{p,i}}\right) \\ \theta_{p,i} &= \mu_p + f_i \\ \mu_p &\sim N(\mu_{pop}, \sigma_p^2) \quad \mu_{pop} \sim N(69.8, 2.6) \quad \sigma_p \sim \text{HalfNormal}(1) \\ f_i &\sim \mathcal{GP}(m(\text{age}_i), k(\text{age}_i, \text{age}_j)) \\ m(\text{age}_i) &= \hat{a} \cdot \text{age}_i^2 + \hat{b} \cdot \text{age}_i \\ k(\text{age}_i, \text{age}_j) &= \eta^2 \exp\left(-\frac{(\text{age}_i - \text{age}_j)^2}{2l^2}\right) \\ \eta &\sim \text{HalfNormal}(0.25) \quad l \sim \text{InvGamma}(15, 14) \\ \sigma_{m,i}^2 &= \begin{cases} \sigma_{obs}^2 & \text{if } y_{p,i} \text{ is observed} \\ \sigma_{imp}^2 & \text{if } y_{p,i} \text{ is imputed} \end{cases} \\ \sigma_{obs} &\sim \text{HalfNormal}(1) \quad \sigma_{imp} \sim \text{HalfNormal}(1)\end{aligned}$$

The parameters $\hat{a}$ and $\hat{b}$ in $m(\text{age}_i)$ were fixed to the posterior estimates fit in Model 1. We parameterized the Inverse-Gamma prior on the lengthscale $l \sim \text{InvGamma}(15, 14)$ to yield a mean of 1.0 with 5th and 95th percentiles of 0.64 and 1.51, respectively. This prior favors smoother trajectories, reflecting the reality that aging effects evolve gradually across multiple seasons rather than fluctuating rapidly year-over-year. The amplitude prior $\eta \sim \text{HalfNormal}(0.25)$ constrains GP deviations around the quadratic mean function $m(\text{age}_i)$, informed by our preliminary quadratic model. For the HSGP approximation, we set $m = 9$ basis functions and a boundary scaling factor of $c = 1.2$, derived using PyMC's approximation helper across a lengthscale range of $[0.75, 1.25]$. Prior realizations from this specification are visualized below:

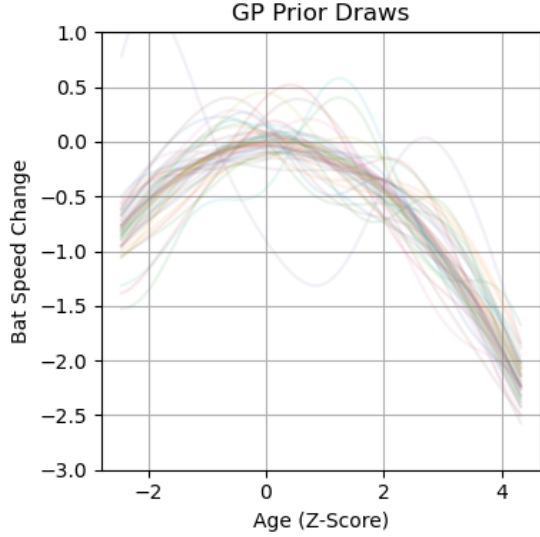


Figure 6: Prior realizations from the HSGP aging curve function

Model 3: HSGP & Position Model

For our final model architecture, we extended the hierarchical structure by incorporating positional offsets into the player random intercepts. The full model specification is defined as follows:

$$\begin{aligned}
y_{p,i} &\sim N\left(\theta_{p,i}, \frac{\sigma_{m,i}^2}{\text{swings}_{p,i}}\right) \\
\theta_{p,i} &= \mu_p + f_i \\
\mu_p &\sim N(\mu_{pop} + \mu_{position[k]}, \sigma_p^2) \quad \mu_{pop} \sim N(69.8, 2.6) \\
\sigma_p &\sim \text{HalfNormal}(1) \\
\mu_{position[k]} &\sim N(0, \sigma_{position}^2) \quad \text{subject to} \quad \sum_{k=1}^K \mu_{position[k]} = 0 \\
\sigma_{position} &\sim \text{HalfNormal}(1) \\
f_i &\sim \mathcal{GP}(m(\text{age}_i), k(\text{age}_i, \text{age}_j)) \\
m(\text{age}_i) &= \hat{a} \cdot \text{age}_i^2 + \hat{b} \cdot \text{age}_i \\
k(\text{age}_i, \text{age}_j) &= \eta^2 \exp\left(-\frac{(\text{age}_i - \text{age}_j)^2}{2l^2}\right) \\
\eta &\sim \text{HalfNormal}(0.25) \quad l \sim \text{InvGamma}(15, 14) \\
\sigma_{m,i}^2 &= \begin{cases} \sigma_{obs}^2 & \text{if } y_{p,i} \text{ is observed} \\ \sigma_{imp}^2 & \text{if } y_{p,i} \text{ is imputed} \end{cases} \\
\sigma_{obs} &\sim \text{HalfNormal}(1) \quad \sigma_{imp} \sim \text{HalfNormal}(1)
\end{aligned}$$

The index k corresponds to the position of player p . To evaluate the effect of positional granularity, we implemented two variants of this structure: Model 3.1 uses standard primary position labels (DH, RF, 3B, SS, C, etc.), while Model 3.2 groups positions into the broader categories (DH, COR, CINF, UTM, and C).

Model Estimation and Validation

To evaluate model performance, we partitioned the dataset chronologically into training and test sets. Observations from 2025 through August 2026 were reserved for out-of-sample evaluation, comprising approximately 18% of the total dataset. A chronological split was required over a standard randomized split to evaluate aging dynamics and prevent data leakage across seasons.

Because the test split is time-based, 194 players in the test set were absent from the training set. Standard frequentist regression approaches would struggle to predict for unobserved players. Under our Bayesian hierarchical structure, unseen players inherit a varying intercept drawn from the posterior distribution of μ_{pop} . Incorporating positional random effects further refines predictions for these unobserved players by anchoring their baseline intercepts to their respective positional prior distributions.

All models were fit using Markov Chain Monte Carlo (MCMC) via the No-U-Turn Sampler (NUTS) implementation in PyMC. Sampling convergence was confirmed across all parameters using $\hat{R} \leq 1.01$ alongside visual inspection of trace plots and posterior predictive checks.

Bat Speed Model Results

We evaluated model fit and out-of-sample predictive accuracy across all candidate specifications. Table 2 details training and out-of-sample performance across all specifications.

Error Metric	Model 1	Model 2	Model 3.1	Model 3.2
Train RMSE	0.782	0.782	0.783	0.783
Test RMSE	2.132	2.133	2.112	2.124
Test Seen RMSE	1.773	1.772	1.769	1.770
Test Unseen RMSE	3.422	3.426	3.357	3.401
Train 95% Coverage	94.4%	94.5%	94.5%	94.5%
Test 95% Coverage	91.1%	91.0%	90.7%	90.7%
Test Seen 95% Coverage	90.9%	90.8%	91.0%	90.9%
Test Unseen 95% Coverage	92.1%	91.9%	89.4%	89.3%

Table 2: Predictive performance across bat speed models

Model 3.1 demonstrated the strongest overall performance, achieving the lowest overall test RMSE, 2.112. Visual inspection of posterior predictive checks confirmed that the model recovers the distribution of observed bat speeds.

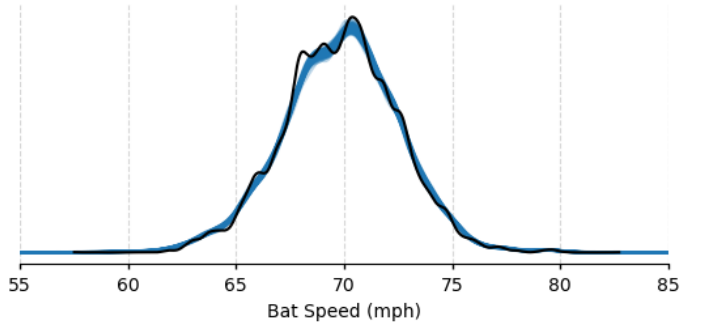


Figure 7: Posterior predictive check

Next we visualize the out of sample performance and identify at which ages the model over or under performs the most. We sized the red dots in the image relative to total observations.

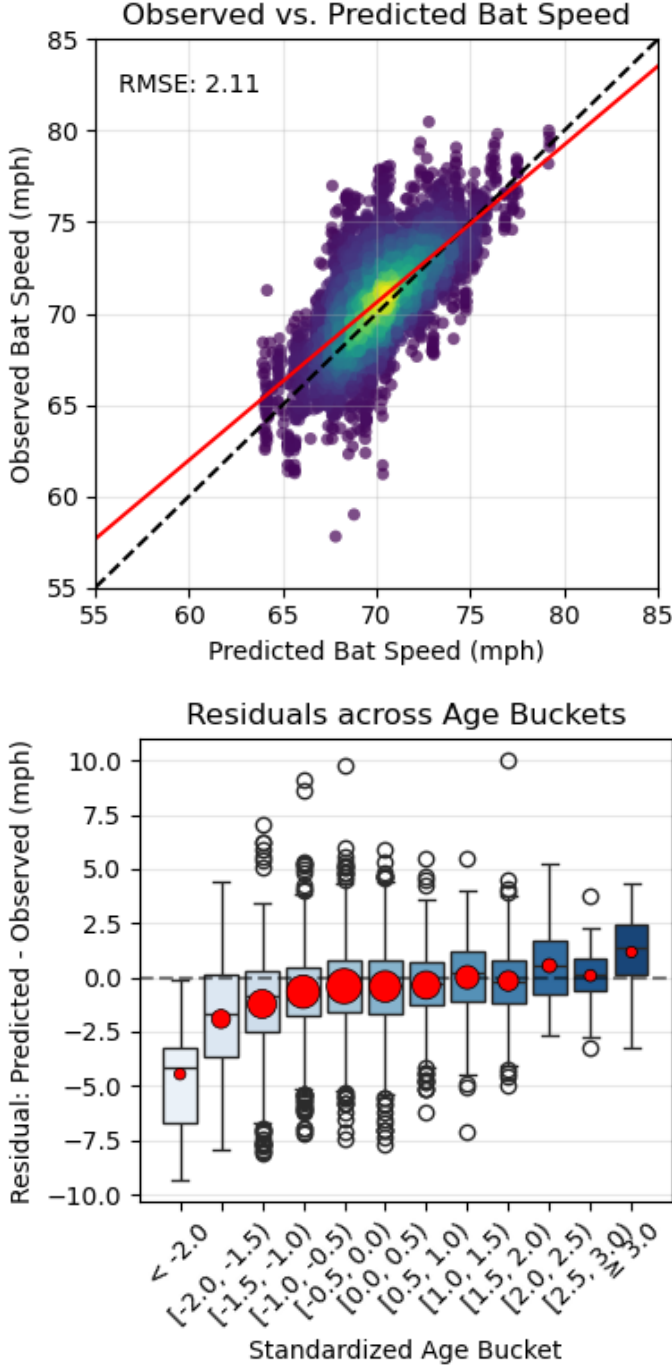


Figure 8: Out of Sample Performance

With Model 3.1 selected as the optimal architecture, we refit the model on the complete 2016-2026 dataset to generate finalized posterior parameters. Fitting the quadratic mean function on the full sample yielded updated coefficients of $\hat{a} \sim \mathcal{N}(-0.13, 0.004)$ and $\hat{b} \sim \mathcal{N}(0.03, 0.01)$, along with a population mean prior of $\mu_{pop} \sim \mathcal{N}(70.0, 2.68)$.

Please see Appendix: Bat Speed Posterior Results for a formal table of posterior summary statistics. The population mean bat speed μ_{pop} achieved a posterior mean of 69.8mph (SD = 0.07). The inter-player standard deviation σ_p was estimated at 2.34mph, quantifying baseline variation across individual hitters. Posteriors for the quadratic mean function coefficients a and b remained tightly centered on their prior

expectations. Variance terms for imputed and observed data yielded posterior means of $\sigma_{imp}^2 = 4.86$ and $\sigma_{obs}^2 = 14.24$, respectively.

For the HSGP aging component, the amplitude parameter η achieved a posterior mean of 0.23 (5th–95th percentile interval: [0.12, 0.42]), while the lengthscale l had a posterior mean of 0.70 ([0.51, 0.92]), which translates into an aging process of 2.6 years given the age distribution. The positional offset variance parameter $\sigma_{position}^2$ had a posterior mean of 1.17. Figure 9 displays the estimated posterior estimations across positions.

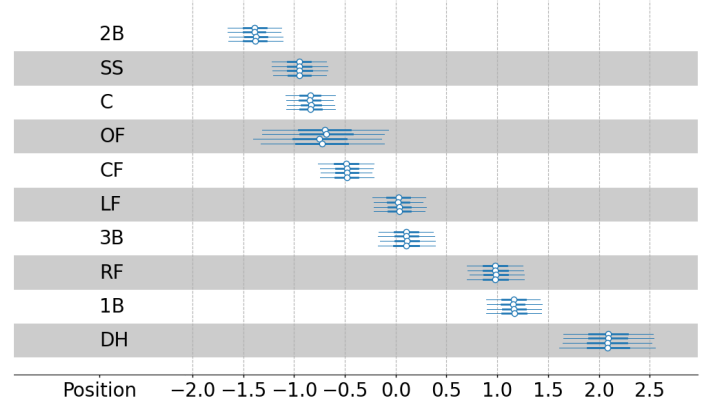


Figure 9: Posterior distributions of positional offsets ($\mu_{position}$)

Finally, Figure 10 illustrates the fitted HSGP population-level aging curve. The corresponding quadratic aging trajectory visualization is provided in Appendix: Quadratic Aging Curve.

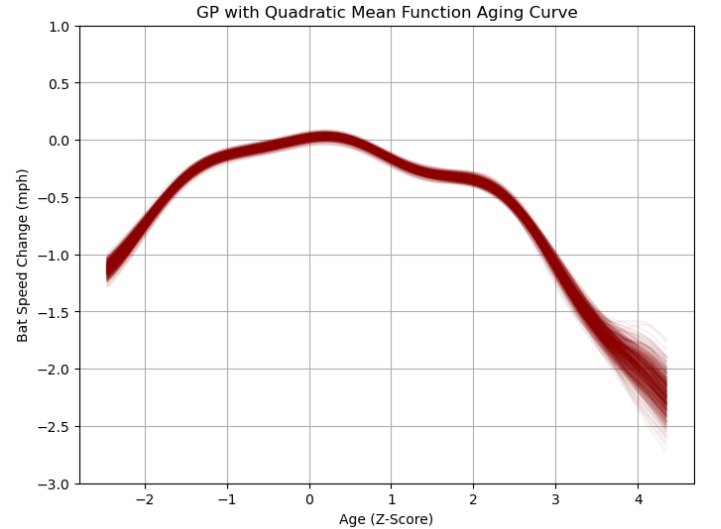


Figure 10: Fitted HSGP aging curve for bat speed

Discussion

Before turning to the full bat speed results, we briefly address validation for our latent exit velocity model and missing data imputation. Because the latent model's primary objective was to estimate true-talent skill rather than generate forward projections, we did not partition this step into training and testing sets. For the missing data imputation, given the sample size constraints and model simplicity, we omitted

out-of-sample validation since there were no hyperparameters to tune. Also, a single-feature setting makes regularization techniques like ridge or lasso regression unnecessary. A spline or even a GP approach would be more flexible and handle the lower tail better. One could have done k-fold cross validation on the coefficient and average the results over those models, but having the posterior distribution is optimal. In short, we can skin this cat a few ways regarding the missing data imputation and spending more time may yield some positive gains.

The estimated positional effects align with domain intuition,

$$\text{DH} > \text{1B} > \text{RF} > \text{3B} > \text{LF} > \text{CF} > \text{OF} > \text{C} > \text{SS} > \text{2B}$$

While one might expect left and right fielders to cluster more tightly, right fielders traditionally profile with higher arm grades to execute throws to third base, a trait that plausibly correlates with greater rotational power and bat speed. The generic “OF” designation presents an interesting effect given that it covers only 30 players in our sample. Thus manually reassigning them to specific spots would be a refinement for future work.

The estimated aging trajectory shows a physical progression. Bat speed builds steadily through a hitter’s late teens and twenties, gaining roughly 1.2 mph from age 19 to a peak near age 28. It then plateaus across the early-to-mid thirties, losing only about 0.4 mph by age 36, before dropping sharply by nearly 2.0 mph over the following eight years. Because public benchmarks for bat speed aging curves do not yet exist, we cannot easily validate whether the true underlying aging effect should exert a larger overall magnitude. For instance, the true curve might exhibit a steeper trajectory with a range closer to a 5–6 mph drop from peak to end rather than our estimated 3mph variation.

Projections

We first display the top and bottom five posterior player effects for hitters who appeared in 2026. We can view these estimated random effects as a player’s underlying true-talent bat speed.

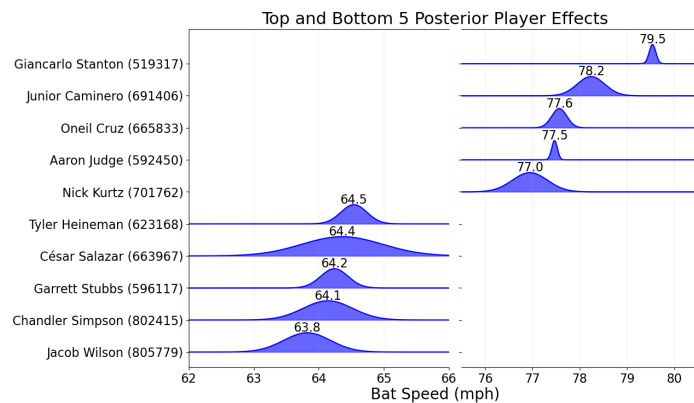


Figure 11: Top and bottom 5 posterior player effects for hitters who appeared in 2026

Extracting these 2026 player effects allows us to generate multi-year projections for several players. First, we examine Nick Kurtz, a young prospect with elite bat speed. Below,

we display his 10-year projection. The underlying Gaussian process projects modest physical development through his mid-twenties, keeping his bat speed in the top tier of the league through age 33. What would be a nice addition is to estimate outcomes given the predicted bat speed, but we save that for future work.

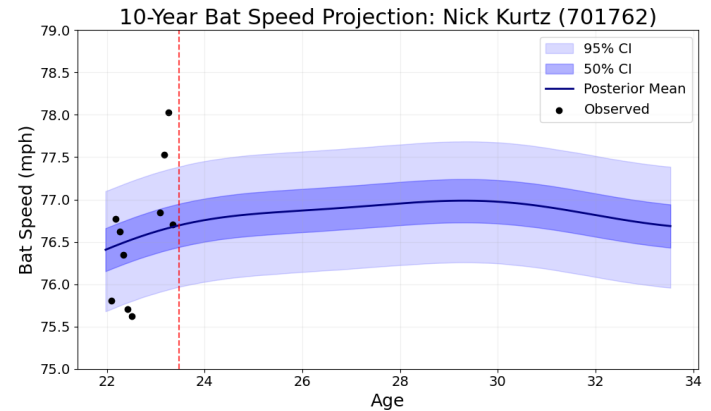


Figure 12: Bat speed projection for Nick Kurtz

We also generate 5-year projections for a broader set of players. Because the population-level GP curve changes bat speed gradually over a player’s prime, individual player random effects primarily dictate overall baseline projection levels, with aging shifts occurring predominantly at the beginning and end of a career.

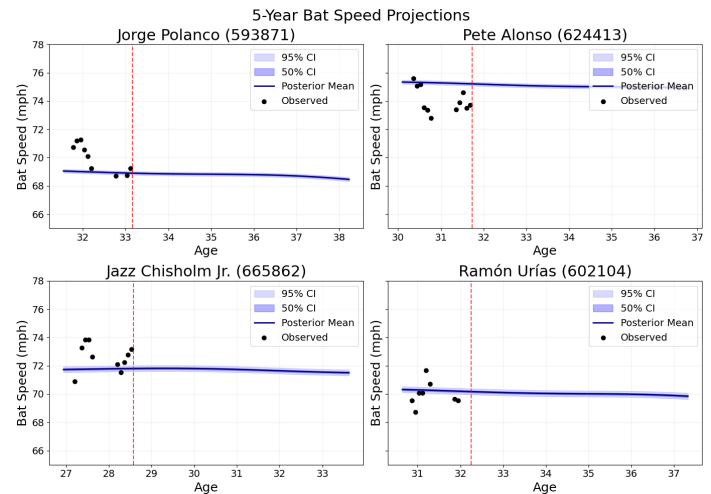


Figure 13: Five-year bat speed projections across select players

Imputed bat speed values heavily drive these estimated player random effects. For example, Luis Arraez posted a low observed bat speed of 62.4 mph, but his imputed bat speed, which draws on a much larger sample size of exit velocity data, was 65.5 mph. This higher imputed value pulled his posterior player effect up to 65.4 mph, causing the model to overestimate his baseline projection. Before adjusting priors or altering the HSGP architecture, focusing on refining the missing data imputation step as suggested earlier will yield the most immediate gains.

Lastly, reviewing model outputs highlighted Jhonkensy Noel as an interesting edge case. Noel posted a posterior player effect in the top 10 across all hitters (76.8 ± 0.52 mph). However,

across 2024 and 2025 with the Guardians, Noel produced a modest .401 career slugging percentage, 19 home runs, and a 32.8% strikeout rate. Thus, while raw bat speed provides a strong foundation for power, it alone does not dictate a hitter’s complete productivity or ultimate career trajectory.

We produced 10Yr projections for every player who appeared in the 2026 season, which can be found here. The code for this project can be found here.

Limitations and Areas for Improvement

Often, researchers gloss over this section. However, normalizing failure is better than ignoring it. The primary area for improvement lies in our missing data imputation given latent exit velocity. Back in 2017, David Marshall laid out steps on FanGraphs to reverse-engineer swing speed at the pitch level [2]. We initially avoided this approach because we wanted to project bat speed rather than spend time on the imputation. However, because our imputation step ignores collision efficiency, incorporating his methodology could prove fruitful.

Second, we know that the population mean of bat speed has increased over time. When we added a time-index effect, the posterior mean of the coefficient sat around 0.01 and did not move the needle for out-of-sample performance. Shifting to a framework that allows underlying talent to evolve over time might help us tease out whether bat speed changes stem from mechanical adjustments or aging.

In terms of technical limitations, tuning our priors and HSGP parameters presented significant hurdles. Including too many basis functions yielded implausible aging curves, while increasing the boundary scaling factor degraded model performance. Additionally, setting a lengthscale prior centered near zero caused convergence failures, forcing us to tighten the prior to enable efficient sampling. We leave a more thorough prior sensitivity analysis for future work. We hypothesize that using monthly observations rather than yearly averages introduced short-term noise that favored shorter lengthscales. Also, assigning a larger prior to η caused convergence issues and generated implausibly large prior draws.

In defense of our framework, the core concept remains straightforward. We assume bat speed evolves over time due to aging. We assign each player an individual baseline via a random effect. For unseen players, we draw from the overall population distribution, apply the corresponding positional offset, and project forward.

Ultimately, our main objectives were to gain hands-on experience with measurement error models, handle missing data through a multivariate framework, and implement an HSGP approximation. We accomplished all three.

Appendix

Bat Speed Model 3.1 Posterior Results

Parameter	Mean	HDI 3%	HDI 97%	ESS Bulk	R-hat
eta	0.23	0.12	0.42	487.73	1.00
ls	0.70	0.51	0.92	811.94	1.00
mu_pop	69.80	69.66	69.95	1544.90	1.00
sigma_player	2.34	2.26	2.43	8222.15	1.00
a	-0.13	-0.14	-0.12	446.30	1.01
b	0.03	0.02	0.05	2148.68	1.00
sigma_imputed	4.86	4.81	4.91	7973.39	1.00
sigma_observed	14.24	14.03	14.46	7807.48	1.00
sigma_position	1.17	0.75	1.84	8644.42	1.00
position_effect[DH]	2.09	1.53	2.64	8735.25	1.00
position_effect[SS]	-0.95	-1.28	-0.61	8908.66	1.00
position_effect[3B]	0.11	-0.23	0.44	9200.04	1.00
position_effect[CF]	-0.48	-0.81	-0.16	7580.02	1.00
position_effect[RF]	0.98	0.65	1.32	8620.70	1.00
position_effect[C]	-0.84	-1.13	-0.55	7986.77	1.00
position_effect[2B]	-1.38	-1.70	-1.06	7509.55	1.00
position_effect[1B]	1.16	0.83	1.49	8929.13	1.00
position_effect[LF]	0.03	-0.28	0.35	8134.73	1.00
position_effect[OF]	-0.71	-1.48	0.04	1807.38	1.01

Quadratic Aging Curve

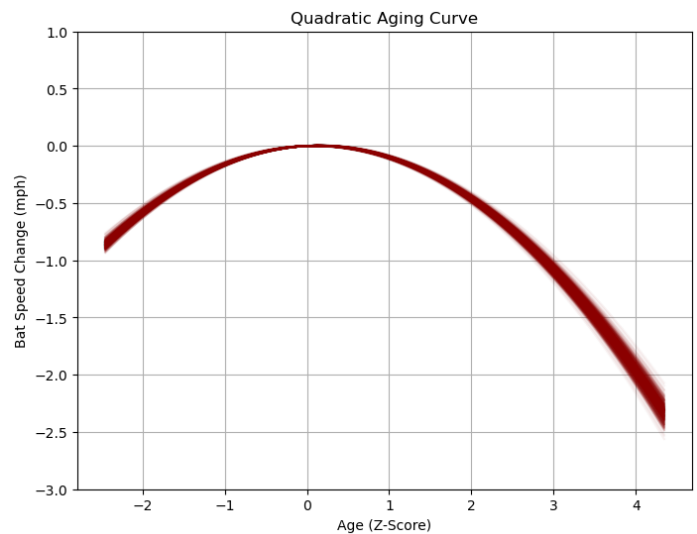


Figure 14: Fitted quadratic aging curve for bat speed

References

- [1] Nathan, Alan M. "Characterizing the Performance of Baseball Bats." American Journal of Physics, vol. 71, no. 2, Feb. 2003, pp. 134–43, <https://baseball.physics.illinois.edu/AJP-Feb2003.pdf>.
- [2] "Reverse-Engineering Swing Mechanics from Statcast Data." FanGraphs Community Research, <https://community.fangraphs.com/reverse-engineering-swing-mechanics-from-statcast-data/>.